

From Diagnosis to a Working System

SLEEP: Final Report — the Localisation Revision and the Four-Model Validation Matrix

Supersedes the 8 August final report; incorporates your localisation feedback and the complete Phase A–D programme

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Professor Guha,

The report of 8 August closed the programme as a diagnosis: a universal recognition–recall dissociation, a universal proxy-validation failure, and a replicated null on the warm-up repair. Your feedback on that version proposed a different reading — that the consolidation failure was a *localisation* error, not a stability–plasticity limit — and three concrete changes: move the consolidation adapter from top-third attention to mid-stack MLP (per the causal-tracing literature), train on paraphrase diversity rather than a single wording, and distil from an in-context teacher. You also asked that adaptation and injection be decoupled, and that knowledge-presence evaluation use greedy decoding.

We implemented all of it, isolated each factor, and then validated the resulting pipeline across the full four-model matrix under pre-registered gates. The one-line summary: **your localisation hypothesis was correct, it was the dominant cause, and the repaired system now beats both naive fine-tuning and an EWC baseline in its design regime.** The recall floor of 0.006 that survived every earlier repair attempt rises to 0.750 on Mistral-7B (five seeds) under the new recipe. This report gives the numbers in the order the evidence was produced.

1. What was run

Stage	Scope executed	Outcome
Ladder (isolation)	5 arms $\times$ 2 models $\times$ 2 seeds	substrate + diversity isolated
Phase A (integration)	recipe wired into the engine; 410 tests	Pass
Phase B (single cycle)	8 runs, machinery A/B	third cause isolated
Phase C (ten cycles)	3 arms $\times$ 2 seeds, Qwen-7B	SLEEP > naive, first time
Phase D (full matrix)	35 runs, 4 models, 5 seeds, EWC baseline	Pass (Llama Flagged)

Phase D ran 11–12 August: 19.7 GPU-hours, $\approx$ USD \$15.5, zero failed runs. Gates and predictions were committed to the repository *before* the matrix launched; every gate was read as written, including the one Llama fails.

2. Your three changes, isolated (the ladder)

Five arms on 50 facts, before touching the full pipeline: the original placement (attention V/O, top third), the new placement (MLP up/down, middle third), each with and without paraphrase

diversity (≥ 20 wordings incl. QA forms), and distillation from an in-context teacher into the new placement. Trained-subset DRA at matched exposure, with background damage (BCP, lower is better):

- **Placement alone halves damage in all 8 of 8 comparisons** (e.g. Qwen BCP 3.6 $\rightarrow$ 1.2–1.8 at equal steps) and lifts recall off the floor.
- **Paraphrases convert memorisation into extraction:** cloze accuracy falls while free-form recall rises — the model stops reproducing a string and starts knowing a fact.
- **Best arms:** Qwen-7B distil $\rightarrow$ mid-MLP 0.36 @ BCP 0.83; Llama paraphrase-CE 0.353. Against the old universal floor of 0.006, the diagnosis was substrate and surface diversity — exactly your reading.

3. The third cause: the safety machinery itself

Integrating the winning recipe into the full autonomous pipeline initially *erased* it: with the safety machinery at its designed setting ($\delta_{\max} = 0.01$ plus plasticity down-scaling), trained-subset DRA is 0.007 across three seeds — indistinguishable from the old floor. Relaxing clip and scaling in the identical pipeline yields 0.210 ± 0.018 at BCP 0.90–0.99, with 21–24 of ~ 50 selected facts passing a direct greedy recall gate — the first externally verified autonomous consolidations in the project. The original failure therefore had three stacked causes: wrong substrate, single-wording training, and safety machinery clipping away the write. A moderate setting ($\delta_{\max} = 0.1$, scaling on) preserves the recall *and* the protection; that setting is used everywhere below.

4. Phase D: the recipe generalises (5 seeds per model)

Model	Recipe	Trained-subset DRA	BCP
Mistral-7B	distil, $\delta_{\max}=0.1$	0.750 ± 0.033	1.21 ± 0.06
Llama-3.1-8B	par.-CE, $\delta_{\max}=0.03$	0.512 ± 0.048	2.09 ± 0.14
Qwen2.5-7B	distil, $\delta_{\max}=0.1$	0.189 ± 0.031	0.88 ± 0.03
Qwen2.5-1.5B	distil, $\delta_{\max}=0.1$	0.105 ± 0.020	1.04 ± 0.02

Every model sits at least $17\times$ above the old floor; the effect is strongest *off* the development model, which argues against overfitting the recipe to Qwen. Distillation beat plain paraphrase-CE wherever both were tried (Mistral: 0.750 @ 1.31 vs 0.655 @ 2.98, seed 0) — the in-context-teacher write is the gentler write. The 1.5B result suggests extraction capacity grows with scale. Llama is the flagged exception: a pre-registered clip ladder ($\delta_{\max} \in \{0.1, 0.05, 0.03\}$) reduced damage monotonically ($4.35 \rightarrow 2.14$) without reaching the ≤ 1.5 bar — its bleed is not clip-dominated, and we report the family as *not yet tuned* rather than excluded.

5. Ten cycles: SLEEP beats naive on all four models

Ten cycles of 20 facts each, cumulative recall at cycle 10 (both seeds shown as s0/s1):

Model	SLEEP v2-moderate	Naive LoRA
Qwen2.5-7B	0.173/0.175 @ 1.87/2.24	0.053/0.060 @ 1.66/1.80
Mistral-7B	0.173/0.168 @ 1.18/1.18	0.118/0.160 @ 1.71/1.57
Qwen2.5-1.5B	0.097/0.102 @ 2.35/2.49	0.038/0.050 @ 24.5/11.4
Llama-3.1-8B	0.102/0.062 @ 4.33/9.20	0.082/0.042 @ 2.24/1.67

Three findings, each consistent across seeds:

1. **The design-intent claim is now real:** SLEEP’s cumulative recall exceeds naive’s on all four models, on both seeds ($3\times$ on Qwen-7B). On Mistral it wins both axes simultaneously — more recall *and* less damage.
2. **The safety machinery is load-bearing at small scale:** naive fine-tuning detonates the 1.5B model (BCP 24.5 and 11.4; the blow-up size is seed-random — the plateau lottery of the 8 August report, reproduced), while SLEEP holds ≈ 2.4 on both seeds.
3. **SLEEP’s signature is stability:** its across-seed spread is ≈ 0.005 on every model, versus naive’s 0.04–0.06. The earlier finding that fixed-horizon comparisons are seed-unreliable still stands — for the *unconstrained* arm. The constrained arm no longer exhibits the lottery.

Llama repeats its single-cycle pattern (recall advantage, unacceptable compounding damage) and keeps its flag.

6. The baseline that matters: EWC-only

The 8 August report noted the uncomfortable finding that EWC alone looked competitive. The matrix settles it with a matched-arms comparison (identical adapter, steps, and learning rate; the only difference is the penalty):

Qwen2.5-7B, cycle 10	SLEEP v2	EWC-only	Naive
Cumulative DRA	0.173/0.175	0.060/0.052	0.053/0.060
BCP	1.87/2.24	1.53/1.80	1.66/1.80

The pre-registered prediction held exactly: EWC preserves (best damage numbers) but its recall sits at naive’s level. **Regularisation prevents forgetting; it does not create extraction.** SLEEP delivers $3\times$ the recall of both baselines at comparable damage. On Llama, EWC-only is competitive with SLEEP (0.075/0.072 at far better BCP) — reported plainly; the Llama column is not a SLEEP win.

7. The paper as it now stands

The paper is being restructured from “anatomy of a failure” into the complete arc: diagnosis (recognition–recall floor, proxy-validation failure, warm-up null — all retained in full) $\rightarrow$ localisation of the cause (your three changes, isolated on the ladder) $\rightarrow$ the third cause (safety machinery) $\rightarrow$ the validated system (the matrix above). The contribution set gains: a causally-decomposed repair of the consolidation failure; cross-family validation at $n=5$ seeds; the EWC dissociation (preservation $\neq$ extraction); and the small-model catastrophe that right-sized machinery prevents. Named open gaps carried into future work: per-family learning-rate tuning for Llama; a background-damage leak

of $\approx +0.12/\text{cycle}$ in all arms; and interleaved rehearsal of prior consolidations (early batches still fade — the mechanism biological sleep actually uses).

8. Recommendation and requested input

Our view: the result is now a positive, multi-family, baseline-controlled finding with pre-registered gates, and the paper should be submitted on that basis. We would welcome your judgement on: (i) whether the diagnosis-era content should remain a full first half or be compressed toward the repair and validation; (ii) the target venue and deadline; and (iii) whether a follow-up programme on interleaved rehearsal — the clearest next mechanism, and the one your multi-cycle instinct pointed at originally — should be scoped now or after review.

Respectfully submitted — Aditya Tripathi. All 35 Phase-D runs, gates, and pre-registrations are in the project repository (`logbook/2026-08-11_phase_d_preregistration.md`, `logbook/2026-08-12_phase_d_full_matrix.md`, results under `experiments/results/pod_run_2026-08-12_phase_d/`). The proposal (7 July), interim report (27 July), and superseded final (8 August) accompany this document.